1 Planning (15 marks)

Defining the problem (3 marks)

P θ is the independent variable or vary θ. [1]

P R is the dependent variable or measure R. [1]

P Keep length of wire constant. [1]

Methods of data collection (5 marks)

M Labelled diagram of apparatus: wire in oil/water bath or oven or beaker with water and source of heat. [1]

M Circuit diagram to measure resistance. [1]

M Use thermometer to measure the temperature of wire/water/oven. (Could be on diagram if labelled.) [1]

M Method to determine resistance from circuit, e.g. read off ohmmeter/$R = V/I$ [1]

M Method to determine R_0 e.g. use ice-water mixture.
Do not allow ice (allow ice at 0 °C or melting ice). [1]

Method of analysis (2 marks)

Do NOT allow log-log graphs.

A	R against θ	R/R_0 against θ	θ against R	[1]
A	α = gradient/R_0	α = gradient	α = 1/ ($R_0 \times$ gradient) α = – 1/y–intercept	[1]

Safety considerations (1 mark)

S Reasoned method to prevent injury from hot water/hot wire e.g. gloves (to prevent injury) from hot water/wire; goggles to prevent splashes from hot water; do not touch hot wire/beaker. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Use insulated wire
2 Use long/thin wire to increase resistance
3 Stir liquid
4 Wait for temperature to stabilise
5 Relationship is valid if straight line, provided plotted graph is correct
6 Relationship is valid if straight line not passing through origin, provided plotted graph is correct (any quoted expression must be correct, e.g. y-intercept = R_0)
7 Use small current to minimise heating effect

Do not allow vague computer methods.

[Total 15]

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